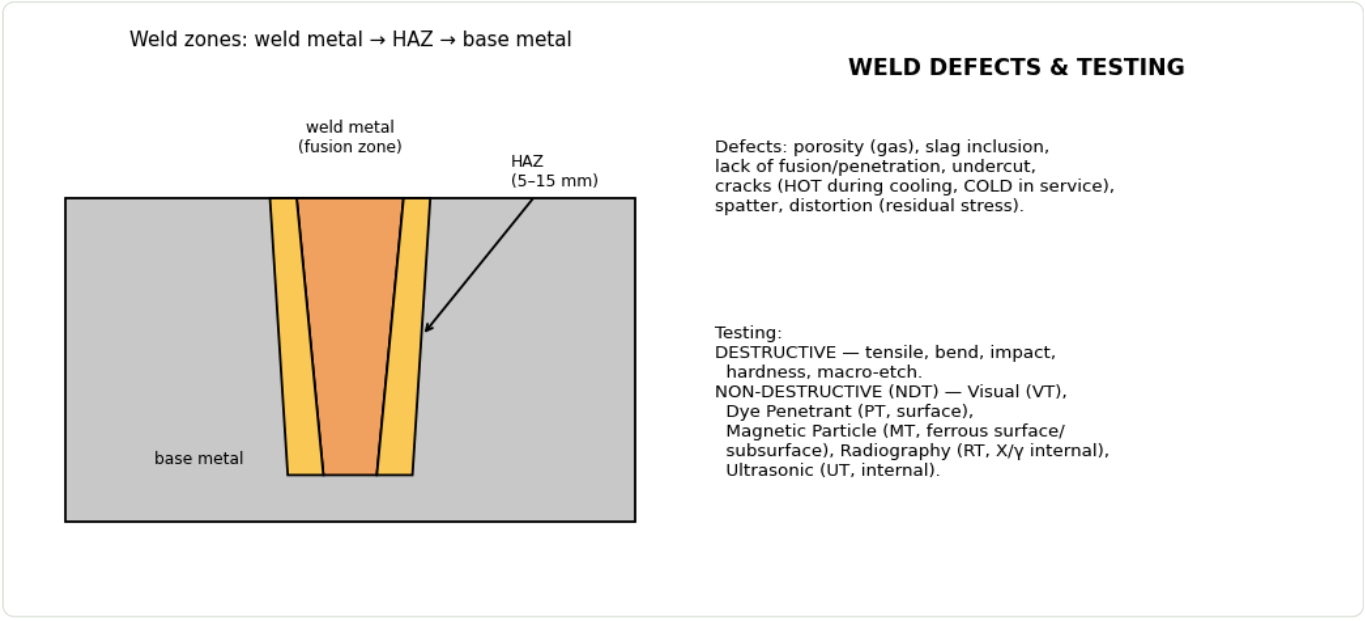
UNIT 8 · TOPIC 6(B)

Weldment Metallurgy, Defects and Testing

Syllabus topic: Basic joining processes: welding, weldment testing and metallurgy

Weld zones and the HAZ



Weld zones (weld metal → HAZ → base metal), defects and testing

Zone	Description
WELD METAL (fusion zone)	Melted and re-solidified — a cast structure
HEAT-AFFECTED ZONE (HAZ)	Base metal NOT melted, but its microstructure/properties were ALTERED by welding heat (grain growth, hardening/softening) — often the WEAKEST, most crack-prone region; typically **5–15 mm** from the fusion line in conventional arc/MIG welds
BASE METAL	Unaffected

HAZ control: limit heat input, preheat, post-weld heat treatment; low-hydrogen electrodes reduce cracking.

Weld defects

Class	Defects
Process / geometry	POROSITY (trapped gas), SLAG INCLUSIONS, LACK OF FUSION / incomplete PENETRATION, UNDERCUT (groove melted into the base metal at the weld toe), overlap, spatter, DISTORTION from residual stress
Metallurgical	HOT cracks (during solidification/cooling); COLD / hydrogen cracks (later, in the HAZ, under service load); lamellar tearing

Causes trace to current/speed, joint preparation, contamination, hydrogen and cooling rate.

Testing of weldments

DESTRUCTIVE tests: tensile, bend (root/face), impact (Charpy), hardness (across the HAZ), macro/micro-etch examination.

NDT method	Detects
VISUAL (VT)	Surface appearance
DYE-PENETRANT (PT)	Surface-breaking flaws — any metal
MAGNETIC PARTICLE (MT)	Surface/near-surface flaws — ferromagnetic metals only
RADIOGRAPHY (RT)	Internal defects by X/γ-rays — permanent film record (also used for castings)
ULTRASONIC (UT)	Internal flaws and thickness, by reflected high-frequency sound

Choose the method by defect type, location and cost (instrumentation — Unit 9).

Practice MCQs

Q92. The base metal region altered by welding heat but not melted is the:

(a) Weld metal (b) Heat-affected zone (HAZ)
(c) Slag (d) Flux

Q93. The HAZ in conventional arc welds typically extends about:

(a) 0.1 mm (b) 5–15 mm
(c) 50 mm (d) 200 mm

Q94. The most crack-prone region of a weldment is often the:

(a) Base metal far away (b) HAZ
(c) Filler spool (d) Flux

Q95. Cracks that occur later in service under load are called:

(a) Hot cracks (b) Cold (hydrogen) cracks
(c) Solidification cracks (d) Craters

Q96. Undercut is a defect that is a:

(a) Gas pore (b) Groove melted into the base metal at the weld toe
(c) Slag pocket (d) Crack

Q97. Distortion in weldments is mainly due to:

(a) Residual (thermal) stresses (b) Painting
(c) Machining (d) Galvanising

Q98. Which is a NON-destructive test?

(a) Tensile test (b) Radiography
(c) Bend test (d) Charpy impact

Q99. Dye-penetrant testing (PT) detects:

(a) Deep internal flaws (b) Surface-breaking defects on any metal
(c) Composition (d) Hardness

Q100. Magnetic-particle testing (MT) is limited to:

(a) Non-magnetic metals (b) Ferromagnetic metals (surface/near-surface)
(c) Plastics (d) Gases

Q101. Internal defects with a permanent film record are best found by:

(a) Visual
(b) Radiography (X/γ-ray)
(c) Penetrant
(d) Hardness

Q102. Ultrasonic testing (UT) detects flaws using:

(a) X-rays
(b) Reflected high-frequency sound
(c) Dye
(d) Magnetism

Answers

Q92 — (b) Heat-affected zone (HAZ) — *HAZ is unmelted but altered.*
Q93 — (b) 5–15 mm — *HAZ ~5–15 mm in conventional arc welds.*
Q94 — (b) HAZ — *Often the weakest, most crack-prone region.*
Q95 — (b) Cold (hydrogen) cracks — *Hot cracks form on cooling; cold/hydrogen cracks appear later in service.*
Q96 — (b) Groove melted into the base metal at the weld toe
Q97 — (a) Residual (thermal) stresses
Q98 — (b) Radiography — *Radiography is non-destructive.*
Q99 — (b) Surface-breaking defects on any metal — *PT — surface flaws, any metal.*
Q100 — (b) Ferromagnetic metals (surface/near-surface) — *MT — ferrous surface/near-surface only.*
Q101 — (b) Radiography (X/γ-ray) — *RT — internal, permanent film.*
Q102 — (b) Reflected high-frequency sound — *UT — internal, reflected sound.*

Sources: Hajra Choudhury — Elements of Workshop Technology (Vol I: casting, welding); Kalpakjian — Manufacturing Engineering; AWS welding references; literature on HAZ, weld defects, NDT and sand vs die casting (as given in the source notes, July 2026); NPTEL manufacturing lectures.